Topic : Introduction to chemistry

S.No.	Sample	Gram Atomic mass of sample	Moles of sample	No. of atoms of sample	Mass removed from the sample	Mole removed	Atoms removed	Mass of same no. of C atom as no. of atoms present in the original sample
1.	8 g **O**	_ _ _	_ _ _	_ _ _	**2 g**	_ _ _	_ _ _	_ _ _
	For Example	**16 g**	$\frac{1}{2}$Mole	$\frac{N_A}{2}$	**2 g**	$\frac{1}{8}$Mole	$\frac{N_A}{8}$	6 g
2.	230 g **Na**				46 g			
3.	60 g **Ca**					1 Mole		
4.	20 g **He**					3 Mole		
5.	56 g **N**					$\frac{1}{2}$Mole		
6.	12 g **Mg**						$\frac{N_A}{4}$	
7.	128 g **S**						N_A	
8.	93 g **P**						$\frac{3N_A}{2}$	

Fill in the blanks (where N_A is Avogadro number) :

S.No.	Mass of sample	Gram molecular mass of sample	Moles of sample	No. of molecules present in the sample	Moles of H-atoms in the sample	No. of C-atoms in the sample	Mass of oxygen in the sample	Total no. of atoms present in the sample
1.	60 g HCHO	_ _ _	_ _ _	_ _ _	_ _ _	_ _ _	_ _ _	_ _ _
	For Example	**30 g**	**2 mole**	**2 N_A**	**4**	**2 N_A**	**32 g**	**8 N_A**
2.			0.5 mole CH_3OH					
3.				3 N_A HCOOH				
4.	8.8 g CH_3CHO							
5.			5 mole C_2H_5OH					
6.				0.25 N_A CH_3COOH				
7.			0.4 mole $C_6H_{12}O_6$					
8.				0.06 N_A $C_{12}H_{22}O_{11}$				

Fill in the blanks (where N_A is Avogadro number) :

S.No.	Gas	Pressure	Volume	Moles of gas in sample	Temperature	Mass of sample	No. of molecules present in the sample	Total no. of atoms present in the sample
1.	NH_3	1 atm	22.4 L		546 K			
2.	CH_4		2.46 L	2 mol	300 K			
3.	O_3	76 torr	4.48 L		1092 K			
4.	N_2	1.013×10^6 pa	0.821 L				0.1 N_A	
5.	CO_2	380 cm of Hg	16.42 L			88 g		
6.	SO_2		100 L		100/0.821 K			0.9 N_A